

LIHUE, HI, USA

WMO: 911650

Lat: 21.984N

Lon: 159.341W

Elev: 100

StdP: 14.64

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22536

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	60.7	62.6	50.8	55.3	67.2	52.9	59.8	67.4	28.3	71.7	26.7	71.4	7.2	280	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	86.1	75.5	85.1	74.9	84.3	74.5	77.7	83.0	76.9	82.4	76.2	81.8	13.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.1	137.0	80.8	75.1	132.1	80.4	74.2	128.2	80.1	41.1	83.3	40.2	82.4	39.5	81.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.9	23.1	20.8	DB	56.8	87.5	2.0	1.8	55.3	88.8	54.1	89.9	53.0	90.9	51.5	92.3
			WB	52.7	78.3	2.2	1.8	51.2	79.6	49.9	80.6	48.7	81.6	47.1	82.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	71.9	71.7	72.5	74.2	75.9	78.0	79.2	79.9	79.7	78.4	75.9	73.5
	DBStd	3.71	2.86	2.59	2.64	2.45	2.26	1.45	1.68	1.70	1.81	1.83	2.33	2.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9463	680	608	696	727	804	840	904	926	891	881	778	728
	CDD65	3989	216	188	231	277	339	390	439	461	441	416	328	263
	CDH74	24486	626	447	636	988	1785	2650	3550	4093	3877	3173	1745	916
	CDH80	3297	15	11	12	32	140	324	592	813	774	441	112	31
Wind	WSAvg	13.0	11.4	12.2	12.9	14.0	12.5	14.0	14.4	13.8	12.7	12.5	13.0	12.5
Precipitation	PrecAvg	37.7	4.2	3.5	4.3	2.9	2.5	1.6	1.9	1.9	2.0	3.7	4.5	4.7
	PrecMax	74.4	14.3	11.3	18.2	10.7	12.2	4.6	6.9	9.9	5.5	18.0	16.1	22.9
	PrecMin	13.5	0.0	0.0	0.2	0.3	0.1	0.3	0.7	0.0	0.4	0.4	0.1	0.1
	PrecStd	12.7	3.8	3.0	3.7	2.2	2.5	1.0	1.1	1.5	1.2	3.1	3.4	4.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.7	81.5	81.4	82.3	83.8	85.0	86.3	87.9	87.9	85.7	83.9	82.7
		MCWB	70.3	69.2	70.9	72.3	73.9	74.2	75.4	77.1	76.6	75.0	74.7	73.2
	2%	DB	80.0	79.4	79.9	80.9	82.5	83.7	84.9	86.2	86.1	84.3	82.4	80.7
		MCWB	70.0	70.1	70.6	71.7	73.2	73.8	74.9	75.7	75.5	74.5	73.6	71.7
	5%	DB	78.6	78.1	78.9	79.9	81.6	82.9	84.1	85.0	84.9	83.4	81.3	79.3
		MCWB	69.7	69.4	70.2	71.2	72.5	73.4	74.4	74.9	74.9	74.2	72.9	71.0
	10%	DB	77.4	76.9	77.6	78.8	80.6	82.1	83.2	84.0	83.9	82.5	80.2	78.2
		MCWB	69.3	68.8	69.5	70.7	71.9	73.1	74.1	74.5	74.4	73.8	72.3	70.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	73.5	74.3	74.7	75.8	76.7	78.2	79.4	78.8	77.5	76.6	75.2
		MCDB	77.1	76.7	78.4	78.9	81.5	81.6	82.5	83.8	83.6	82.8	81.1	79.6
	2%	WB	72.4	72.3	72.5	73.3	74.8	75.5	77.0	77.7	77.5	76.4	75.5	73.8
		MCDB	76.7	76.2	77.0	78.5	80.5	81.2	82.2	83.5	82.7	81.9	80.1	78.1
	5%	WB	71.4	71.2	71.4	72.4	73.7	74.8	76.1	76.7	76.5	75.6	74.6	72.9
		MCDB	76.3	75.7	76.3	77.8	79.4	80.6	81.6	82.6	82.2	81.3	79.4	77.3
	10%	WB	70.6	70.2	70.5	71.5	72.9	74.0	75.3	75.8	75.7	74.9	73.5	71.8
		MCDB	75.7	75.1	75.7	77.0	78.7	79.9	81.0	81.8	81.8	80.8	78.4	76.5
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.3	10.1	8.9	9.5	8.7	8.8	8.8	9.3	9.1	8.8	9.6
		MCDBR	11.8	11.8	11.0	9.4	9.7	9.1	8.9	9.2	9.7	9.4	9.4	10.2
		MCWBR	6.5	6.6	5.9	4.9	4.6	4.1	4.1	4.1	4.3	4.5	5.0	5.6
	5% WB	MCDBR	10.6	9.8	10.1	9.0	9.4	8.6	8.4	8.9	9.2	9.2	9.0	9.4
		MCWBR	6.6	6.4	6.0	5.2	4.9	4.4	4.6	4.4	4.7	4.8	5.2	5.6
Clear-Sky Solar Irradiance	taub	0.353	0.356	0.362	0.394	0.385	0.372	0.373	0.370	0.364	0.367	0.361	0.347	
	taud	2.608	2.588	2.577	2.473	2.527	2.571	2.577	2.585	2.609	2.615	2.625	2.650	
	Ebn at Noon	292	297	299	289	288	289	289	292	294	290	287	291	
	Edh at Noon	28	30	32	36	34	32	32	32	30	29	28	26	
All-Sky Solar Radiation	RadAvg	1397	1560	1809	2062	2211	2308	2281	2187	2036	1690	1422	1280	
	RadStd	57	127	195	119	129	60	58	85	60	65	44	91	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page